

OpenCV

1. Otwarcie obrazu

```
img = imread("zdjecie.jpg", cv2.IMREAD_COLOR)
```

https://docs.opencv.org/4.x/db/deb/tutorial_display_image.html

2. Wyświetlenie obrazu

```
imshow("Display window", img);  
waitKey(0);  
cv2.destroyAllWindows()
```

https://docs.opencv.org/4.x/db/deb/tutorial_display_image.html

3. Piksele

```
red[:,2]=0
```

https://docs.opencv.org/3.4/d3/df2/tutorial_py_basic_ops.html

4. Histogram

```
grayscale_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)  
hist_grey = cv2.calcHist([grayscale_image], [0], None, [256], [0, 256])
```

https://docs.opencv.org/3.4/d1/db7/tutorial_py_histogram_begins.html

5. Progowanie

```
_, thresh = cv2.threshold(imgray, 140, 255, 0)
```

https://docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html

6. Segmentacja K-średnich

```
ret,label,center=cv.kmeans(Z,K,None,criteria,10,cv.KMEANS_RANDOM_CENTERS)
```

https://docs.opencv.org/3.4/d1/d5c/tutorial_py_kmeans_opencv.html

7. Segmentacja kolorem

```
light_black = (0, 0, 200) #kolor HSV  
dark_black = (145, 60, 255) #kolor HSV  
mask_black = cv2.inRange(hsv_img, light_black, dark_black)  
result_black = cv2.bitwise_and(img, img, mask=mask_black)
```

<https://realpython.com/python-opencv-color-spaces/>

8. Filtracja

```
kernel = np.array([[1, 1, 1],  
                  [1, 1, 1],  
                  [1, 1, 1]])  
  
kernel = kernel/(np.sum(kernel) if np.sum(kernel)!=0 else 1)  
  
result2 = cv2.filter2D(image,-1,kernel)
```

https://docs.opencv.org/3.4/d4/dbd/tutorial_filter_2d.html

9. Inne Filtry

a. Medianowy

```
medianBlur(img, 7)
```

https://docs.opencv.org/4.x/d4/d13/tutorial_py_filtering.html

b. Gauss

```
GaussianBlur(maskedImage, (5, 5), 0)
```

https://docs.opencv.org/4.x/d4/d13/tutorial_py_filtering.html

c. Canny

```
Canny(blur,50,150)
```

https://docs.opencv.org/3.4/da/d22/tutorial_py_canny.html

10. Rozpoznawanie

a. Rozpoznawanie konturów

```
contours, hierarchy = cv2.findContours(thresh, cv2.RETR_TREE,  
cv2.CHAIN_APPROX_SIMPLE)
```

https://docs.opencv.org/3.4/d4/d73/tutorial_py_contours_begin.html

b. Aproksymacja konturów

```
approxPolyDP(cnt, 0.01*cv2.arcLength(cnt, True), True)
```

https://docs.opencv.org/3.4/dc/dcf/tutorial_js_contour_features.html

c. Wykrywanie kształtów

<https://pyimagesearch.com/2016/02/08/opencv-shape-detection/>